

## Detection searches for the APT29 attack emulation

Table 1: SPL queries linked to the steps in the APT29 attack scenario

| Emulation plan step  | Detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| PowerShell           | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where match(CommandLower, "powershell\.exe")      AND (like(CommandLower, "%-windowstyle hidden%") OR like(CommandLower, "%-w hidden%"))    stats values(CommandLine) as "Command Lines" by ComputerName </pre>                                                                                                                          |
| Automated Collection | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1 "powershell"    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where match(CommandLower, "powershell\.exe")    rex field=CommandLower max_match=0 "(? &lt;raw_matches&gt;\.doc \.xps \.xls \.ppt \.pps \.wps \.wpd \.ods  \.odt \.lwp \.jtd \.pdf \.zip \.rar \.docx \.url \.xlsx \.pptx \.ppsx \. pst \.ost \.psw \.bpass \.blogin \.badmin \.sifr sifer  bvpn \.jpg \.txt \.lnk)" </pre> |

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|                                                           | <pre>   eval distinct_types = mvdedup(raw_matches)    where mvcount(distinct_types) &gt;= 3    stats values(CommandLine) as "Command Lines", values(distinct_types) as "Unique Indicators" by ComputerName </pre>                                                                                                                                            |
| System Owner / User<br>Discovery                          | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where like(CommandLower, "%echo %userdomain%\\ %username%%")    stats values(CommandLine) as "Command Lines" by ComputerName </pre>  |
| Data from staged file and<br>Exfiltration over C2 Channel | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where like(CommandLower, "%invoke- multipartformdataupload%")    stats values(CommandLine) as "Command Lines" by ComputerName </pre> |
| Process Discovery                                         | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" </pre>                                                                                                                                                                                                                                                            |

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|                   | <pre> EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where like(CommandLower, "%import-module%")        AND like(CommandLower, "%processlist%")    stats values(CommandLine) as "Command Lines" by ComputerName </pre>                                                                                                 |
| Account Discovery | <pre> index=sysmon sourcetype="WinEventLog"  LogName="Microsoft-Windows-Sysmon/Operational"  EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where (like(CommandLower, "%net%user%/domain%") OR like(CommandLower, "%net1%user%/domain%"))    stats values(CommandLine) as "Command Lines" count by ComputerName </pre> |
| Account Discovery | <pre> index=sysmon sourcetype="WinEventLog"  LogName="Microsoft-Windows-Sysmon/Operational"  EventCode=1    rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"    eval CommandLower=lower(CommandLine)    where (like(CommandLower, "%net%user%/domain%") OR like(CommandLower, "%net1%user%/domain%")) </pre>                                                                       |

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|                               | AND NOT CommandLower LIKE "%/add%"<br><br>AND NOT CommandLower LIKE "%/del%"<br><br>  stats values(CommandLine) as "Targeted User Domain Lookups" count by ComputerName                                                                                                                                                                                                                                                                                                                       |
| Staging monkey PNG            | index=sysmon<br><br>sourcetype="WinEventLog"<br><br>LogName="Microsoft-Windows-Sysmon/Operational"<br><br>EventCode=1<br><br>  eval CommandLower=lower(_raw)<br><br>  where (like(CommandLower, "%system.drawing%"))<br><br>OR (<br>(like(CommandLower, "%copy%") OR<br>like(CommandLower, "%move%"))<br>AND like(CommandLower, "%.png%")<br>AND NOT like(CommandLower, "%remove%")<br>)<br><br>  stats values(CommandLine) as "Command Lines" count by ComputerName<br><br>  where count > 1 |
| UAC Bypass via Backup Utility | index=sysmon<br><br>sourcetype="WinEventLog"<br><br>LogName="Microsoft-Windows-Sysmon/Operational"<br><br>EventCode=1<br><br>  eval CommandLower=lower(CommandLine)                                                                                                                                                                                                                                                                                                                           |

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|                                           | <pre>   where match(CommandLower, "powershell") AND match(CommandLower, "hkcu")   rex field=CommandLower max_match=0 "(?&lt;_exes&gt;[a- zA-Z0-9\-\.\_]+\.\exe)"   eval other_exe = mvfilter(NOT match(_exes, "powershell\.\exe pwsh\.\exe"))   where isnotnull(other_exe)   table _time, host, User, Image, other_exe, CommandLine, ParentImage   sort - _time </pre>                                                                       |
| Registry Cleanup for UAC Bypass Technique | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" (EventCode=1 OR EventCode=12)   eval CommandLower = lower(CommandLine), TargetLower = lower(TargetObject)   where (match(CommandLower, "remove-item") AND match(CommandLower, "hkcu")) OR (EventCode=12 AND match(TargetLower, "hkey_current_user"))   table _time, host, User, EventCode, Image, CommandLine, TargetObject   sort - _time </pre> |
| Process Injection                         | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" </pre>                                                                                                                                                                                                                                                                                                                                            |

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|                                             | <pre> EventCode=1    eval CommandLower=lower(CommandLine)    where match(CommandLower, "import-module")      AND match(CommandLower, "tobase64string")      AND match(CommandLower, "start-job")      AND match(CommandLower, "\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}")    table _time, host, User, Image, CommandLine    sort - _time </pre>                                                                             |
| Planting Modified<br>Sysinternals Utilities | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1    eval cmd=lower(CommandLine)    where (like(cmd, "%iwr%") OR like(cmd, "%invoke- webrequest%")) AND like(cmd, "%copy-item%")    table _time, host, User, Image, CommandLine </pre>                                                                                                                           |
| Loading Stage-2 &<br>Performing Discovery   | <pre> index=sysmon LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1    where (match(CommandLine, "(?i)(Move-Item mv move  ren).*\. (txt log csv tmp dat bak old).*\.ps1"))      AND (match(CommandLine, "(?i)\.ls+\.\\.+\.ps1") OR match(CommandLine, "Invoke-"))    eval PotentialScriptSource = case(      match(CommandLine, "\.txt"), "Text File",      match(CommandLine, "\.log"), "Log File", </pre> |

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|                                                                        | <pre> match(CommandLine, "\.csv"), "CSV File", match(CommandLine, "\.tmp"), "Temp File", 1=1, "Other Extension" )   rename CommandLine as cmd   table _time, Computer, User, PotentialScriptSource, cmd </pre>                                                                                                                              |
| Credential Dumping using Process Injection                             | <pre> index=sysmon LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1   eval cmd=lower(CommandLine)   where (     match(cmd, "lsadump::sam") AND     match(cmd, "iwr wget curl webclient downloadstring  downloadfile") AND     match(cmd, "\biex\b invoke-expression") )   table _time, host, User, CommandLine </pre>             |
| Data from staged file (T1074) and Exfiltration over C2 Channel (T1041) | <pre> index=sysmon sourcetype="WinEventLog" LogName="Microsoft-Windows-Sysmon/Operational" EventCode=1   rex field=Message "CommandLine:\s+(? &lt;CommandLine&gt;[^\r\n]+)"   eval CommandLower=lower(CommandLine)   where like(CommandLower, "%invoke- multipartformdataupload%")   stats values(CommandLine) as "Command Lines" by </pre> |

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|                                                                              | ComputerName                                                                                                                                                                                                                                                                                                                                                     |
| Copy Sandcat File                                                            | index=sysmon<br>LogName="Microsoft-Windows-Sysmon/Operational"<br>EventCode=1<br>  eval cmd=lower(CommandLine)<br>  search cmd="*psexec*" AND (cmd="*move*" OR<br>cmd="*mv*" OR cmd="*mi*" OR cmd="*copy*" OR<br>cmd="*cp*") AND cmd="*.exe*"<br>  where match(cmd, "\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}")<br>  table _time, host, User, Image, CommandLine       |
| Data from staged file<br>(T1074) and Exfiltration over<br>C2 Channel (T1041) | index=sysmon sourcetype="WinEventLog"<br>LogName="Microsoft-Windows-Sysmon/Operational"<br>EventCode=1<br>  rex field=Message "CommandLine:\s+(?<br><CommandLine>[^\r\n]+)"<br>  eval CommandLower=lower(CommandLine)<br>  where like(CommandLower, "%invoke-<br>multipartformdataupload%")<br>  stats values(CommandLine) as "Command Lines" by<br>ComputerName |